

Equation of a Straight Line

Straight-Line Graphs. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner connects the equation $y = mx + c$ to graphs, tables and real-world stories: reading the gradient as a rate of change and the y-intercept as a starting value, interpreting positive, negative and zero gradients, writing equations, and explaining what a linear model means in context.

Driving Question: *Can a line tell the story of change?*

The mathematics, in plain terms

A straight line can be written $y = mx + c$. Here m is the gradient — how much y changes each time x increases by one — and c is the y-intercept, the value of y when $x = 0$ (the starting value). A positive gradient rises left to right, a negative gradient falls, and a zero gradient is flat. In a real model, m is a rate (for example, cost per person) and c is a starting amount. Reading, writing and interpreting these two numbers is the whole lesson.

Two numbers, two jobs

Almost every difficulty here comes from mixing up the two numbers or mishandling their signs. It helps to keep their jobs separate and concrete: m answers “how fast, and in which direction?”, c answers “where does it start?”. With negative values, encourage reading the sign as part of the number (a gradient of -2 means y goes *down* by 2 each step). Linking each number to a real story — a rate and a starting amount — makes both stick.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Equation, table, graph and story are four views of the same line — each connected to the others.	~15 min
Movement	Changing m and c and watching the line move — steeper, gentler, up, down.	~12 min
Reflection	Asking: which number is the rate, which is the start? What does the sign tell me?	~7 min
Creativity	Writing a human story that a given line could tell.	~4 min
Rest	The pause to predict how the line will move before changing a number.	~3 min

Nutrition	Intellectual (Mode B): feeds the ability to read change and make sense of real trends and choices.	~4 min
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Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Confusing the gradient and the intercept

Reading c as the gradient or m as the start. Repair: “Which number multiplies x ? That is the rate. Which number stands alone? That is where it starts.”

Sign errors with negative gradients or intercepts

Repair: “Is the line rising or falling? A falling line has a negative gradient — read the minus sign as part of the number.”

Treating gradient as steepness only

Ignoring direction. Repair: “Two lines can be equally steep but one rises and one falls. What does the sign of the gradient tell you?”

Losing the meaning in a real model

Repair: “In this story, what does the gradient measure? What does the starting value mean when nothing has happened yet ($x = 0$)?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: negative numbers, the idea that every point on the y -axis has $x = 0$, and substitution. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the two numbers, and their signs, in view:

- “Which number multiplies x ? What does it tell you about how fast y changes?”

- “Which number is on its own? Where does the line cross the y-axis?”
- “Is the line rising or falling — and does the sign of the gradient agree?”
- “In this story, what does the gradient mean, and what does the starting value mean?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“They keep swapping m and c .”

Anchor each to its job: m multiplies x (the rate of change); c stands alone (the starting value, where $x = 0$). Linking both to a real story helps them stay put.

“Negative gradients confuse them.”

Encourage reading the minus as part of the number: a gradient of -3 means y goes down by 3 for every step of 1 in x . Checking against a rising or falling line confirms it.

“Do they need to plot perfectly by hand?”

Understanding what m and c *mean* matters more than neat plotting. A sketch that shows the right start and direction is strong evidence.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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